

Basic Statistical Analysis

Pablo E. Gutiérrez-Fonseca, PhD

University of Puerto Rico

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Basic Statistical analysis

- Now we are going to cover how to perform a variety of basic statistical tests in R.

1. T-tests

2. Correlation

3. Linear Regression

4. One-way ANOVA

Basic Statistical analysis

- Now we are going to cover how to perform a variety of basic statistical tests in R.
1. T-tests (independent and dependent)
 2. Correlation (Pearson, Spearman, Kendall)
 3. Linear Regression (Simple, Multiple, Logistic,)
 4. One-way ANOVA (Factorial ANOVA, Full Factorial ANOVA, Repeated Measures ANOVA)

t-tests

t-tests

When would you run a t-test?

1. You have a continuous response variable.
2. You want to test for a **difference between two sample means**: considering the variance within each group, do the groups belong to two distinct populations, or not?
3. Data is **normally distributed** and both groups have approximately **equal variance**.
4. If the groups are **not related** in any way = **Independent t-test**
5. If the groups are **related (paired)** = **Dependent t-test** (or matched pairs t-test).

Independent vs. Dependent groups

- How do I know if my samples are independent or dependent?
- Can you tell exactly which observation should be paired with which other observation? If you can't tell, your groups are independent

Independent vs. Dependent groups

Examples of Independent

Samples 1. Treated and untreated, randomly located plots.

2. Monitoring sites set up with Systematic (grid) sampling.

3. Infested and uninfested forest stands selected with stratified random sampling.

Examples of Dependent

Samples 1. Before and After data taken on the same observation.

2. Paired watersheds / plots.

3. Parent / offspring studies.

4. Twin studies / partner studies.

t-tests

- The T-test is performed using the `t.test()` function, which essentially tests for the difference in **means of a variable between two groups**. It can perform a one sample t-test, two sample t-test, and a paired t-test.
- In this syntax `t.test(x, y)`, x and y are the columns of data for each group. It performs a two sample t-test assuming unequal variances.

```
1 t.test(x= , y = ,  
2       mu = 0,  
3       paired = ,  
4       `var.equal`= `,  
5       alternative = c("two.sided", "less", "greater")  
6       )
```

Example: Two-tailed t-test

Research Question: The state is looking to choose a stream for habitat restoration for native trout species. Streams with deeper pools will provide necessary cover in the winter months and minimize potential scouring from severe rain events. Which stream should they focus on? Test to see if there is a significant difference in depth between stream 1 and stream 2?

```
1 stream_data <- read.csv('data/t_test_two_tail_water_data.csv')
2 head(stream_data)
```

	X	waterbody	depth_cm
1	1	stream1	42.6
2	2	stream1	48.4
3	3	stream1	46.2
4	4	stream1	44.2
5	5	stream1	44.4
6	6	stream1	50.6

Step 1. Normality test

- Test for normality

```
1 byf.shapiro(depth_cm~waterbody, data=stream_data)
```

Shapiro-Wilk normality tests

data: depth_cm by waterbody

	W	p-value
stream1	0.9698	0.4749
stream2	0.9819	0.7822

Step 2. Equal variance

```
1 #var.test  
2 var.test(depth_cm ~ waterbody, data=stream_data)
```

F test to compare two variances

data: depth_cm by waterbody

F = 0.99478, num df = 32, denom df = 37, p-value = 0.9945

alternative hypothesis: true ratio of variances is not equal to 1

95 percent confidence interval:

0.5080442 1.9810856

sample estimates:

ratio of variances

0.9947842

Step 3: Conducting the t-test

```
1 t.test(filter(stream_data, waterbody == "stream1")$depth_cm,  
2         filter(stream_data, waterbody == "stream2")$depth_cm,  
3         alternative = "two.sided",  
4         mu = 0,  
5         paired = FALSE,  
6         var.equal = TRUE)
```

Two Sample t-test

data: filter(stream_data, waterbody == "stream1")\$depth_cm and filter(stream_data, waterbody == "stream2")\$depth_cm

t = -12.924, df = 69, p-value < 0.000000000000000022

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

-13.298846 -9.742143

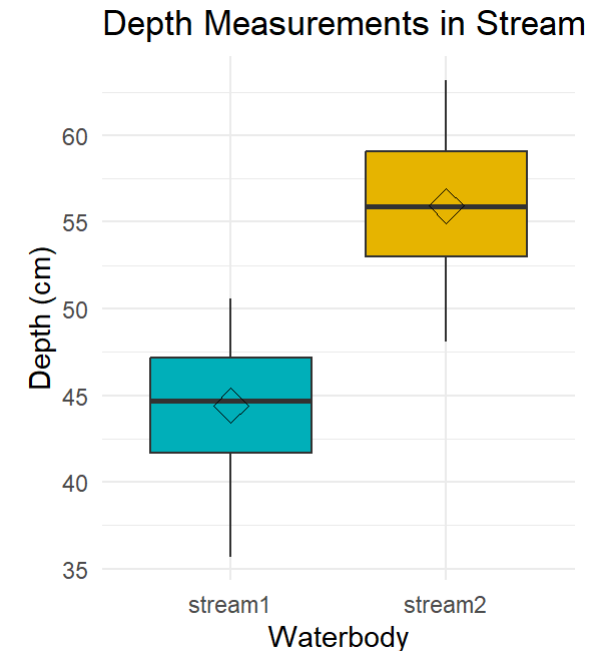
sample estimates:

mean of x mean of y

44.42424 55.94474

Visualization

```
1 ggplot(stream_data, aes(x = waterbody,  
2                       y = depth_cm,  
3                       fill = waterbo  
4 geom_boxplot() +  
5 stat_summary(fun = mean, geom = "poi  
6             shape = 5, size = 4, co  
7 labs(title = "Depth Measurements in  
8       x = "Waterbody",  
9       y = "Depth (cm)") +  
10 theme_minimal() +  
11 scale_fill_manual(values =  
12                   c("#00AFBB", "#E  
13 theme(legend.position = "none")
```



Correlation

What is correlation?

- A correlation test quantifies how two variables change together.
 - Are the variables closely related?
 - To what extent does one variable predict the other?

When to use correlation?

- Correlation test is used to **evaluate the association between two or more variables.**
- You have two continuous variables measured from the same observational unit.
- You want to determine if the variables are related (vary in unison).

Key assumptions:

- Independent, random samples.
- Normality:
 - Pearson correlation (parametric): Assumes both variables follow a normal distribution.
 - Spearman correlation (nonparametric): Does not assume normality.
- Linear relationship: Correlation looks for a linear association between the variables.

Describing correlations

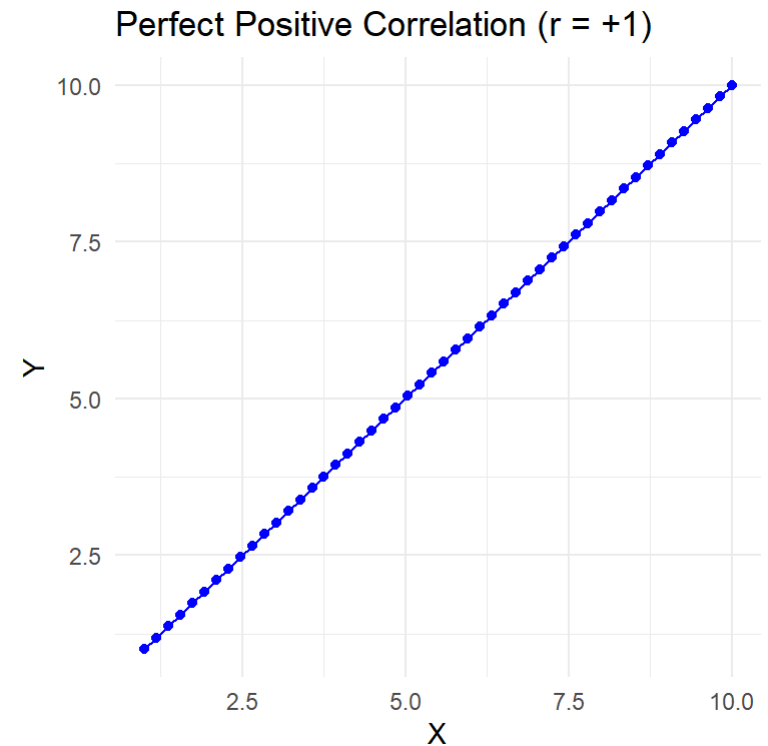
Correlation in R

- `cor()` performs correlation in R

```
1 cor(x, y = NULL, use = "everything",  
2     method = c("pearson", "kendall", "spearman"))
```

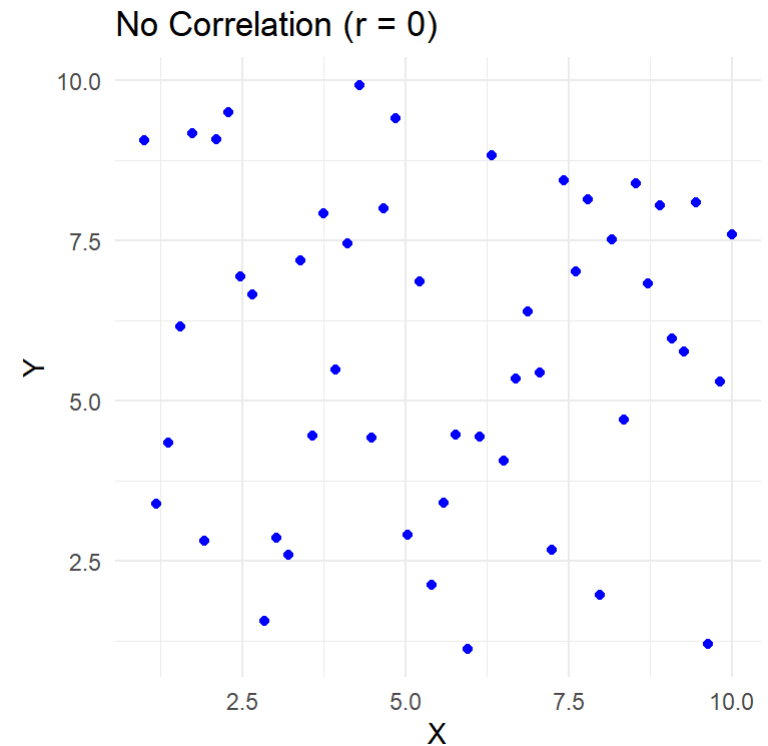
Nature and Interpretation of r Value

- **Perfect Positive Correlation:**
 $r = +1$
 - All points lie on an upward-sloping line.



Nature and Interpretation of r Value

- Perfect Positive Correlation:
 $r = +1$
 - All points lie on an upward-sloping line.
- No Correlation: $r = 0$
 - Points do not align along any line, or they form a horizontal line.

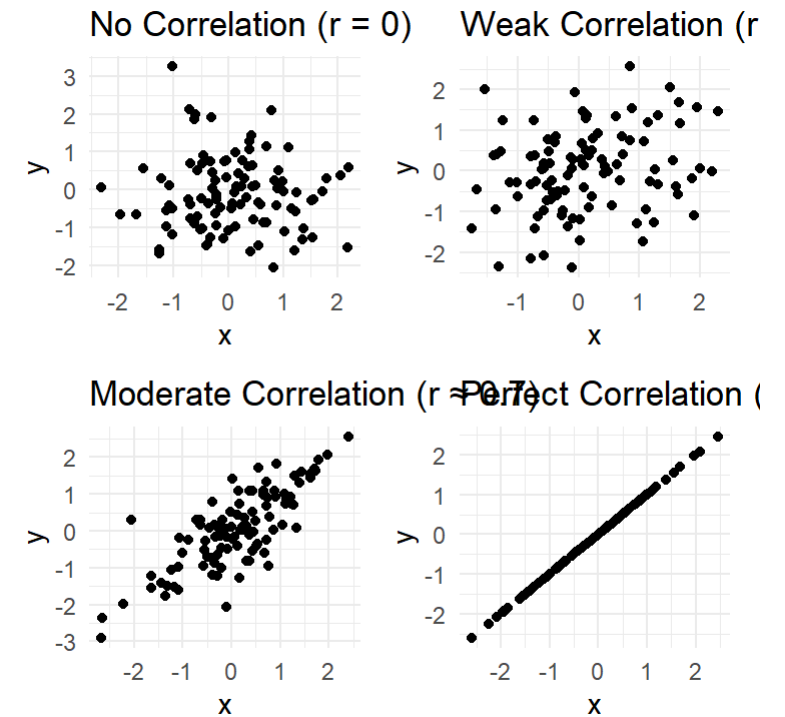


Nature and Interpretation of r Value

Nature and Interpretation of r Value

Strength of Correlation

- Weak Correlation:
 - $0.0 \leq |r| < 0.3$
- Moderate Correlation:
 - $0.3 \leq |r| < 0.7$
- Strong Correlation:
 - $0.7 \leq |r| < 0.9$
- Perfect Correlation:
 - $|r| = 1.0$



Coefficient of Determination (r^2)

- r^2 quantifies the proportion of variability in one variable that can be explained by its relationship with the other variable.
- It represents the shared variability between the two variables, indicating how much of the total variability they have in common.
- When two variables are correlated, r^2 tells us how much variance in one variable can be accounted for by the other.
- This concept is similar to the Effect Size we examined in ANOVA.

Difference between r and r^2

- r (Pearson's Correlation Coefficient):
 - Measures how much two variables vary together.
 - Ranges from -1 to 1:
 - -1: Perfect negative correlation
 - 0: No correlation
 - 1: Perfect positive correlation
 - Unitless and applicable across contexts.

Difference between r and r^2

Example: Correlation

Relationship between CO₂ and Global Temperature

Load Data: Import the dataset containing global temperature and CO₂ concentration.

```
1 global_T <- read_excel('data/correlation global mean T.xlsx')
2 tail(global_T)
```

A tibble: 6 × 3

	year	temperature	Mona_Loa_co2
	<dbl>	<dbl>	<dbl>
1	2013	14.6	397.
2	2014	14.7	399.
3	2015	14.8	401.
4	2016	14.9	404.
5	2017	14.8	407.
6	2018	14.7	409.

Relationship between CO₂ and Temperature

Test for Normality: Use Shapiro-Wilk tests to check if temperature and Mona_Loa_co2 are normally distributed.

```
1 shapiro.test(global_T$temperature)
```

Shapiro-Wilk normality test

```
data:  global_T$temperature  
W = 0.9314, p-value = 0.000002771
```

Relationship between CO₂ and Temperature

Test for Normality: Use Shapiro-Wilk tests to check if temperature and Mona_Loa_co2 are normally distributed.

```
1 shapiro.test(global_T$Mona_Loa_co2)
```

Shapiro-Wilk normality test

```
data:  global_T$Mona_Loa_co2  
W = 0.94085, p-value = 0.005914
```


Relationship between CO₂ and Temperature

This code performs a **Spearman correlation** test to examine the relationship between global temperature and CO₂ concentration.

```
1 #Test correlation (parametric)
2 test <- cor.test(global_T$temperature,
3                 global_T$Mona_Loa_co2,
4                 method = "spearman")
5 test
```

Spearman's rank correlation rho

data: global_T\$temperature and global_T\$Mona_Loa_co2

S = 2403.6, p-value < 0.000000000000000022

alternative hypothesis: true rho is not equal to 0

sample estimates:

<https://pegutierrezf.github.io/TropicalFreshwaterEcology/>

Coefficient of Determination (R^2) for Spearman Correlation

Approximate the proportion of variance in temperature explained by CO₂ concentration.

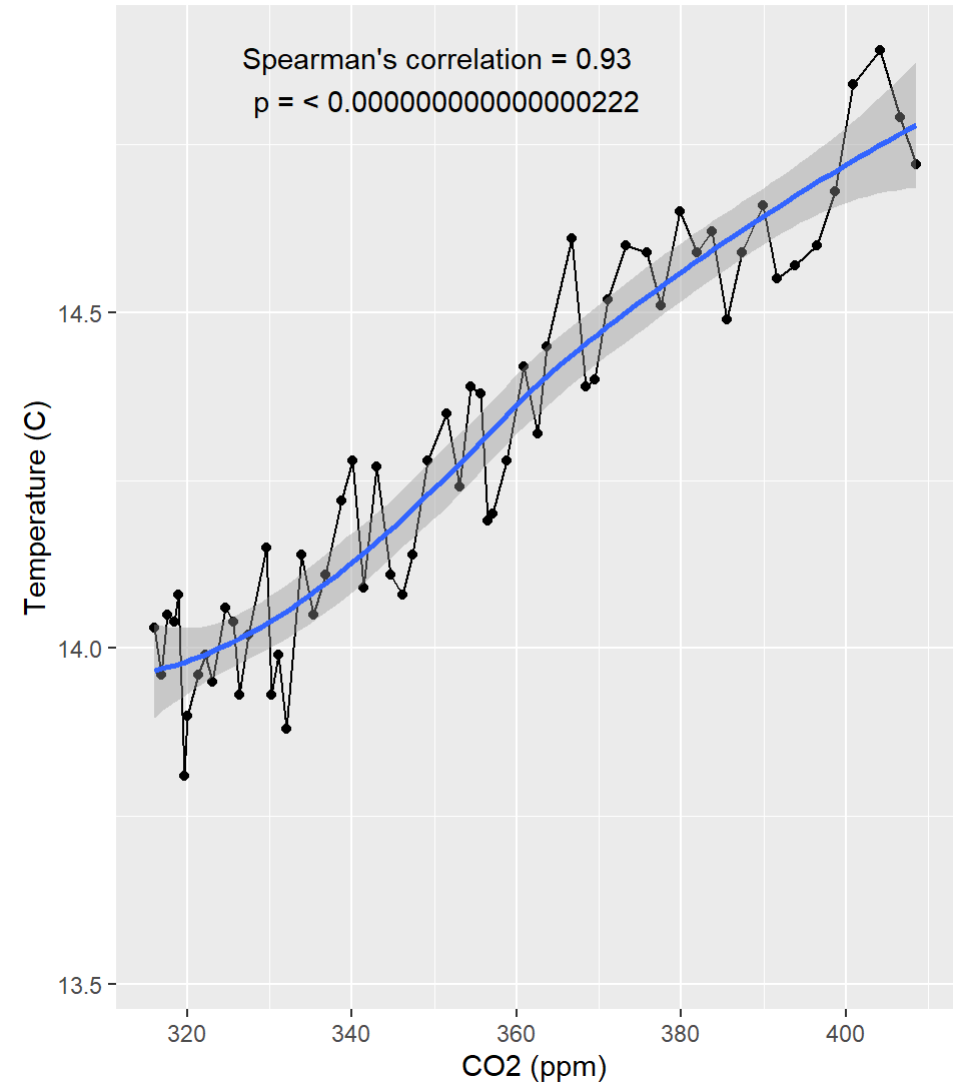
```
1 rho <- test$estimate # Spearman correlation coefficient
2 R_squared <- rho^2
3 R_squared
```

```
rho
0.8708898
```

```
1 # Extract correlation coefficient and p-value
2 p_value <- test$p.value
```

Data visualization

```
1 ggplot(global_T,
2         aes(x = Mona_Loa_co2,
3             y = temperature)) +
4   geom_line() +
5   geom_point() +
6   geom_smooth() +
7   # ylim(14,14.75) +
8   labs(x = "CO2 (ppm)",
9        y = "Temperature (C)",
10        title = "") +
11   annotate("text", x = 375, y = 14.8,
12              label = paste("Spearman's rho = ",
13                            round(rho, 2), "\n",
14                            "p = ", format.pval(p_value),
15                            hjust = 1, vjust = 0))
```



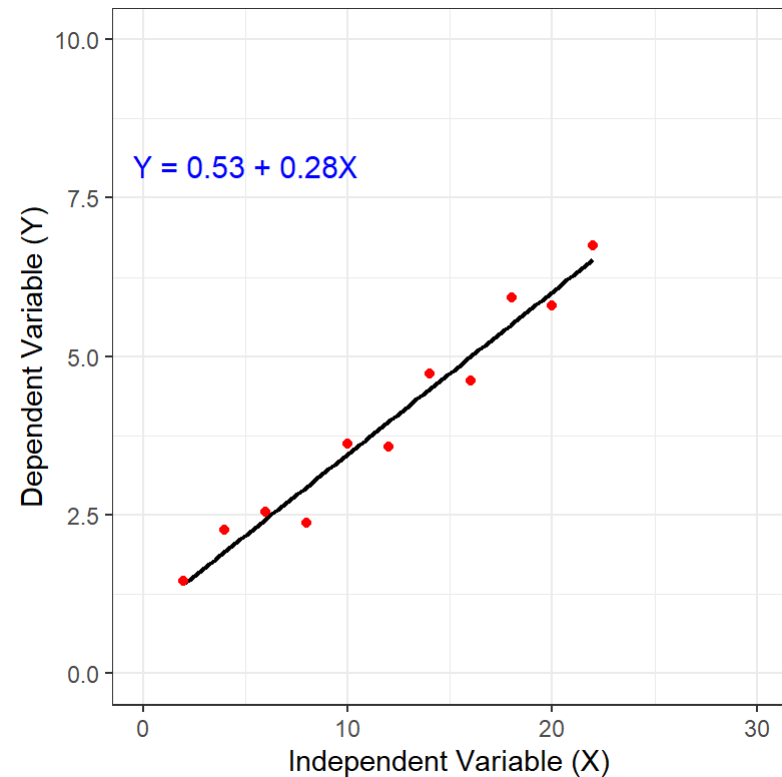
Linear Regression

What is regression?

- Regression analysis is a generic term for a group of different statistical techniques.
- The purpose of all these techniques is to examine the relationship between variables.
 - Regression aims to find the **best-fitting linear equation** ($Y = \beta_0 + \beta_1 X_1 + \epsilon$) that describes the relationship between the dependent variable (often denoted as Y) and independent variables (denoted as $X_1, X_2, \dots, X_n$).

What can we accomplish with linear regression?

1. Describing the nature of the relationship between two variables.



What can we accomplish with linear regression?

What can we accomplish with linear regression?

When to use a Simple Linear Regression

- You have two continuous variables measured in the same unit.
- You suspect the variables are related and vary in unison.
- You aim to predict one variable based on the other.
- You want to evaluate a trend (slope) over multiple measurements.

Assumptions

Mathematically

- The simple linear model is defined as:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Mathematically

- The multiple linear model is defined as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

Linear Models in R

- `lm()` fits linear models in R
 - Simple linear regression

```
1 lm(Y ~ X, data = dataset)
```

- where:
 - y is your outcome
 - x is your predictor

Linear Models in R

- `lm()` fits linear models in R
 - Multiple linear regression

```
1 lm(Y ~ X1 + X2 + X3, data = dataset)
```

- where:
 - y is your outcome
 - x is/are your predictor(s)

Example: Simple linear regression

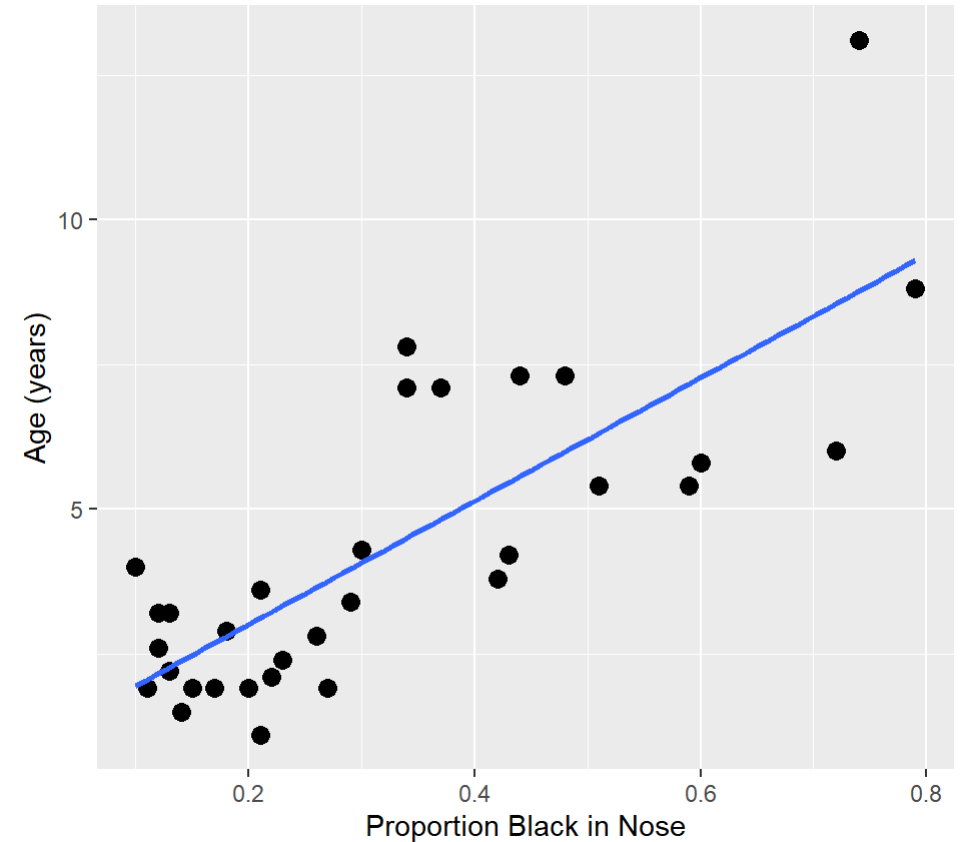
```
1 lion_data <- read_excel("data/SLR_lion.xlsx")
2 head(lion_data)
```

A tibble: 6 × 2

	proportionBlack	ageInYears
	<dbl>	<dbl>
1	0.21	1.1
2	0.14	1.5
3	0.11	1.9
4	0.13	2.2
5	0.12	2.6
6	0.13	3.2

Step 1. Visualize the data

```
1 p <- ggplot(lion_data,  
2             aes(x = proportionBlack,  
3                 y = ageInYears)) +  
4   geom_point(size=3) +  
5   xlab('Proportion Black in Nose') +  
6   ylab('Age (years)') +  
7   geom_smooth(method='lm',  
8               se=F)  
9 p
```



Step 2. Fit the linear model

```
1 mod <- lm(ageInYears~proportionBlack, data = lion_data)
2 summary(mod)
```

Call:

```
lm(formula = ageInYears ~ proportionBlack, data = lion_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.5449	-1.1117	-0.5285	0.9635	4.3421

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.8790	0.5688	1.545	0.133
proportionBlack	10.6471	1.5095	7.053	0.0000000768 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Step 3. Check model assumptions:

Normality

- Get list of residuals to test normality

```
1 res <- resid(mod)
2 shapiro.test(res)
```

Shapiro-Wilk normality test

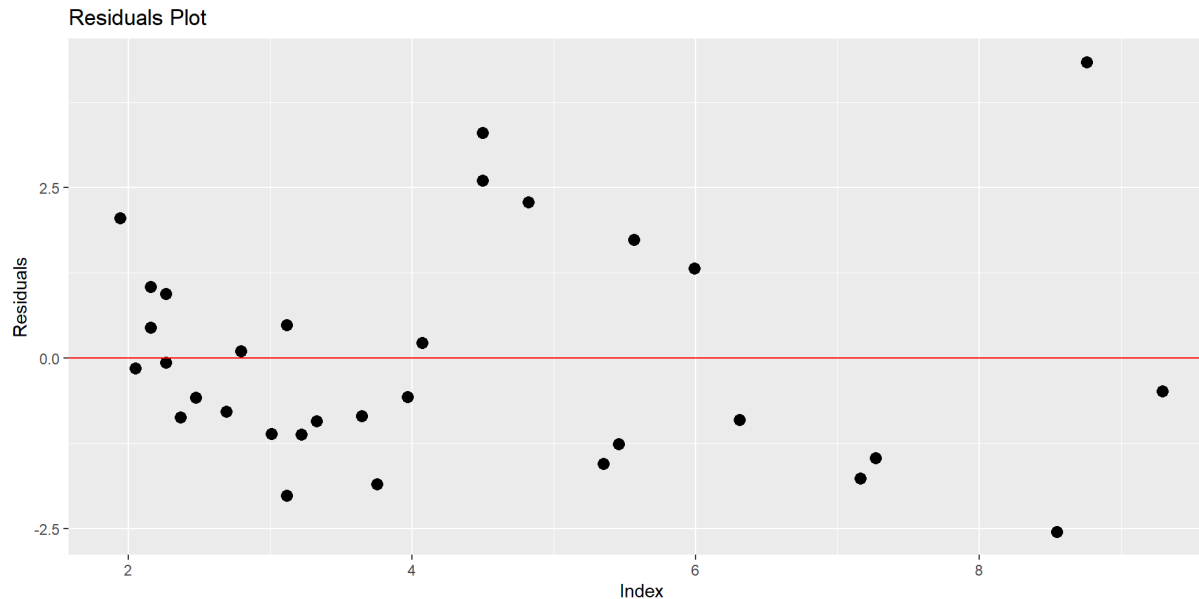
data: res

W = 0.93879, p-value = 0.0692

Residuals plot

- Create residual plot

```
1 ggplot(mod, aes(x = .fitted, y = .resid)) +  
2   geom_point(size=3) +  
3   geom_hline(yintercept = 0, color = "red") +  
4   labs(title = "Residuals Plot",  
5         x = "Index",  
6         y = "Residuals")
```



Analysis Of Variance (ANOVA)

What is ANOVA?

- ANOVA (Analysis of Variance) is a statistical technique used to analyze the relationship between categorical variables and continuous variables in R.
- This statistical method is an extension of the independent samples t-test for comparing the means in a situation where **there are more than two groups.**

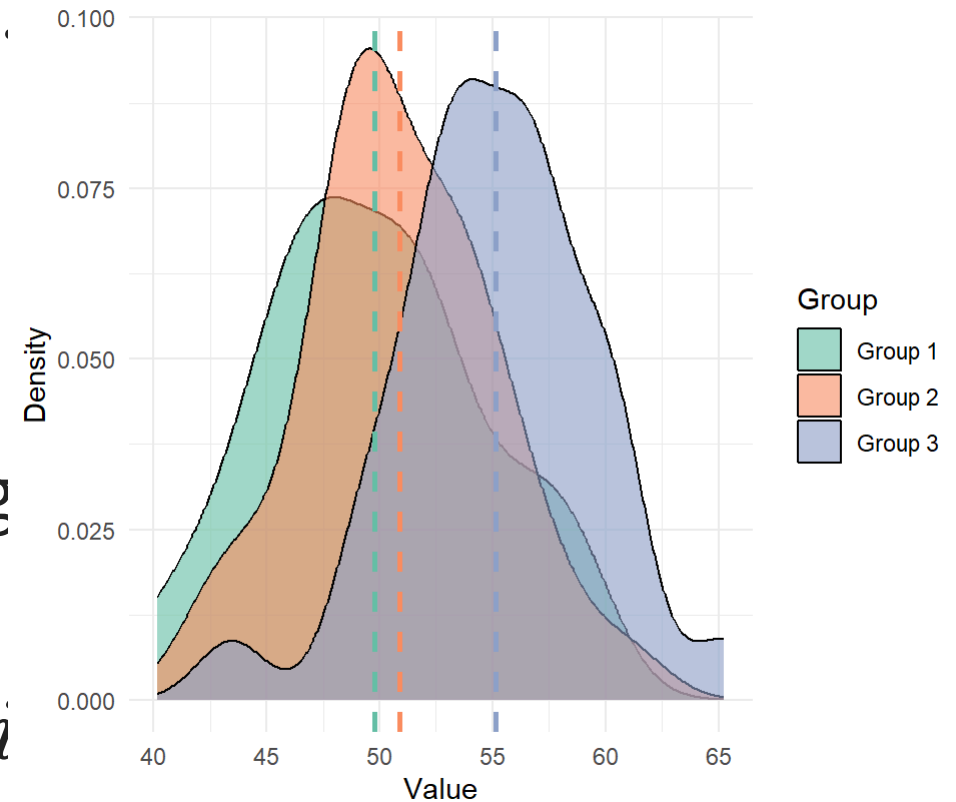
Hypothesis in one-way ANOVA test

- H_0 : All group means are equal.

- $H_0 : \mu_1 = \mu_2 = \mu_3 = \dots$

- H_a : At least, the mean of one group is different

- $H_A : \mu_i \neq \mu_j$ for some i



When would you run an ANOVA?

- I have a continuous response variable.
- I want to know the difference in that response between more than two groups.
- I have one factor I am testing (One-Way ANOVA).

Assumptions

- There are three key assumptions that you need to be aware of:
 1. All groups are Independent.
 2. The data is normally distributed.
 3. All groups have approximately equal variance.
 - A good rule of thumb: ratio of largest to smallest group stdev must be less than 2:1
 - The ANOVA is less sensitive to this requirement when samples are of equal size from each population.

Assumptions

- A common misconception is that the **response variable** must be normally distributed when conducting an ANOVA. This is **incorrect** because the normality assumptions pertain to the **residuals**, not the response variable.

Assumptions

How ANOVA works

How ANOVA works

- The ANOVA obtained value (the **F-statistic**) is a ratio of the Between Group Variation divided by the Within Group Variation:

$$F = \frac{\text{Between-group variance}}{\text{Within-group variance}}$$

- A large **F** is evidence against H_0 , since it indicates that there is more difference between groups than within groups.

One-way ANOVA Formulas

Example: One-way ANOVA

```
1 crop <- read.csv('data/ANOVA_crop.csv')
2 head(crop)
```

	density	block	fertilizer	yield
1	1	1	ctrl	177.2287
2	2	2	ctrl	177.5500
3	1	3	ctrl	176.4085
4	2	4	ctrl	177.7036
5	1	1	ctrl	177.1255
6	2	2	ctrl	176.7783

Data visualization

- Visualize the data distribution using boxplots and jitter for individual points:

Running the ANOVA

- Run one-way ANOVA to assess differences between fertilizer groups:

```
1 one.way <- aov(yield ~ fertilizer,  
2               data = crop, var.equal = TRUE )  
3  
4 summary(one.way)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
fertilizer	2	6.07	3.0340	7.863	0.0007 ***
Residuals	93	35.89	0.3859		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Checking Assumptions: Normality

Checking Assumptions: Normality

Check Assumptions: Homogeneity of Variances

The homogeneity of variances can also be checked using **Bartlett's**.

```
1 bartlett.test(yield ~ fertilizer, data=crop)
```

Bartlett test of homogeneity of variances

data: yield by fertilizer

Bartlett's K-squared = 1.0622, df = 2, p-value = 0.5879

Post-hoc test

Why Perform a Post-hoc Test?

- When the **overall p-value** from the ANOVA is below the significance level (typically $p < 0.05$), it suggests that **at least one group mean is different from the others**.
- However, ANOVA does not specify which groups are different; it only tells us that **not all group means are equal**.
- To pinpoint which groups differ, we need to conduct a **post-hoc test**. This test controls for the family-wise error rate, ensuring accurate comparisons between group means.